MODEL ASSESSMENT AND RECOMMENDATIONS

STADIOalot: Predicting E-commerce Product Returns

To: STADIOalot Management

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Executive summary

STADIOalot's product-return rate rose from 22% in 2025 to 29% in 2026, increasing pressure on margins and logistics. Of the proposed models, Random Forest is the stronger candidate, outperforming Logistic Regression on every reported test metric. Its 62.90% accuracy and 0.6844 ROC-AUC nevertheless indicate that improvement and local validation are needed before deployment.

1. Best model: Model 1 or Model 2

Metric	Logistic Regression	Random Forest
Accuracy	57.80%	62.90%
Precision	53.73%	59.73%
Recall	56.98%	58.38%
F1-score	55.31%	59.05%
ROC-AUC	0.6088	0.6844
Cross-validation ROC-AUC	0.6062	0.6649

Random Forest is recommended because it improves accuracy by 5.10 percentage points, precision by 6.00 points, recall by 1.40 points and F1-score by 3.74 points. Its higher ROC-AUC indicates better ranking of likely returns across thresholds (scikit-learn, n.d.). The confusion matrix supports this choice: Random Forest correctly classified 7,229 non-returns and 5,350 returns, compared with 6,339 and 5,222 for Logistic Regression. It produced fewer false positives (3,607 versus 4,497) and false negatives (3,814 versus 3,942).

Its main gain is avoiding false alarms: it correctly identifies 890 more non-returns but only 128 more returns. Random Forest is therefore the best model tested, not yet a production-ready solution. Both models should be compared on identical test observations to ensure a fair evaluation.

2. Improving Random Forest

Broaden hyperparameter tuning beyond the six configurations tested in Part C. Test tree numbers, maximum depth, minimum leaf size and feature sampling through cross-validation, selecting settings on validation data and retaining a separate test set.

Engineer features such as prior customer/product return rates, delivery delays, review ratings or sentiment, and return patterns by reason. Sharma (2024) engineered delivery time and customer/product return rates and found XGBoost achieved 76.9% accuracy among seven classifiers. This supports testing these features and benchmarking XGBoost, but the score is not directly comparable because the dataset and class balance differ.

Prevent data leakage by using only information available at the prediction time. Return reason, refund method, return date and final return status are recorded after a return and should not predict a pre-return outcome. Use IDs to link records rather than as standalone signals, and calculate historical rates only from earlier transactions.

Finally, tune the classification threshold to business costs rather than automatically using 0.50. A lower threshold may catch more returns but create more false alarms; a higher threshold may miss returns. Assess precision, recall, F1-score and confusion matrices alongside handling costs and customer experience. Pilot interventions such as clearer descriptions or size guidance, then measure whether they actually reduce avoidable returns.

3. Adapting the model to SS1 data

SS1 specifies five linked datasets: historical sales returns, customer reviews, delivery, inventory and return policies (Data type table eCommerce and logistics Capstone project, n.d.). They should be joined into an order-item-level modelling table.

SS1 dataset	Proposed use and key caution
Sales returns	Use Order ID, Product ID and Customer ID to link outcomes and calculate historical rates. Return-only data lacks non-returned purchases; obtain the full sales/order population.
Customer reviews	Use earlier star ratings, verified-buyer status or

	review-text sentiment. Exclude reviews posted after prediction.
Delivery	Use courier or delay features only when known at prediction time; actual delivery date and final status are future information for checkout predictions.
Inventory	Use stock, warehouse and cost context. Time-stamped snapshots are needed to reconstruct what was known then.
Return policies	Join category, return window and policy version effective on the order date; do not use later policy versions.

Validate unique IDs, dates, controlled categories, numeric values, missing data and table links according to the SS1 data dictionary. Define the target as whether each purchased item is returned within a suitable observation window. Label an item as a non-return only after that window has elapsed, or recent orders may be mislabelled.

Do not apply the current model unchanged: SS1 has different tables and fields. Map features, exclude post-event information, retrain on labelled STADIOalot data and compare Logistic Regression, Random Forest and XGBoost using consistent splits. A chronological test period is useful for predicting future returns; report accuracy, precision, recall, F1-score, ROC-AUC and confusion matrices.

4. Alignment with Part A literature

The results align with Part A's emphasis on comparing algorithms and engineering useful features. Sharma (2024) supports testing delivery and historical return-rate features and benchmarking XGBoost. Bothe (2020) describes return prediction using online clothing-retailer data and Random Forest/XGBoost methods, supporting tree-based comparisons; however, the dataset was artificially balanced, limiting direct comparison with STADIOalot.

Bivera and Mohan (2025) discuss a hybrid approach combining cart-level deep learning and item-level gradient boosting and report 86% accuracy. As noted in the Part A review, the evaluation details do not clearly establish a directly comparable combined-model test, so this figure should not be treated as

STADIOalot's target. Published results differ in datasets, class balance and evaluation metrics; accuracy figures cannot be compared directly with STADIOalot's ROC-AUC. Random Forest's better performance is consistent with the literature, but its moderate results show the need for local testing. SS1's extra data may improve prediction only if it is reliable and available at the correct time.

Conclusion

Select Random Forest as the leading candidate, broaden tuning, engineer time-safe features, benchmark XGBoost and set the threshold according to business costs. Join SS1 datasets to full sales/order records, prevent leakage and validate on future-period data. A controlled pilot should confirm whether predictions reduce avoidable returns and improve margins without harming customer experience.

References

Bivera, J. and Mohan, C.V. (2025) 'Product return prediction in e-commerce platforms', International Research Journal on Advanced Engineering and Management, 3(12), pp. 3292–3299. Available at: https://www.researchgate.net/publication/399110482_Product_Return_Prediction_in_E-Commerce_Platforms (Accessed: 05 October 2026).

Bothe, A.F. (2020) Order Returns Prediction Dataset. GitHub. Available at: https://github.com/AnFrBo/order_returns_prediction (Accessed: 05 October 2026).

Data type table eCommerce and logistics Capstone project (n.d.) SS1 data dictionaries.

Sharma, N. (2024) Analyzing Customer Behavior Patterns & Predicting Online Product Return Intentions: A Data Mining Approach. Master's thesis. Rhine-Waal University of Applied Sciences. Available at: https://journals.mmupress.com/index.php/jiwe/article/view/1638 (Accessed: 05 October)